As an example of using the analysis equations, we will find the frequency response of the RC low-pass filter. This is done by taking the Fourier transform of its impulse response, previously shown in Fig. 13-4, and described by:

$$h(t) = 0 \qquad \text{for } t < 0$$

$$h(t) = \alpha e^{-\alpha t} \qquad \text{for } t \geq 0$$

The frequency response is found by plugging the impulse response into the analysis equations. First, the real part:

$$Re\,H(\omega) = \int_{-\infty}^{+\infty} h(t)\cos(\omega t)\,dt \qquad \text{(start with Eq. 13-3)}$$

$$Re\,H(\omega) = \int_{0}^{+\infty} \alpha e^{-\alpha t}\cos(\omega t)\,dt \qquad \text{(plug in the signal)}$$

$$Re\,H(\omega) = \frac{\alpha e^{-\alpha t}}{\alpha^2 + \omega^2}\left[-\alpha\cos(\omega t) + \omega\sin(\omega t)\right]\Bigg|_{0}^{+\infty} \qquad \text{(evaluate)}$$

$$Re\,H(\omega) = \frac{\alpha^2}{\alpha^2 + \omega^2}$$

Using this same approach, the imaginary part of the frequency response is calculated to be:

$$Im\,H(\omega) = \frac{-\omega\alpha}{\alpha^2 + \omega^2}$$

Just as with discrete signals, the rectangular representation of the frequency domain is great for mathematical manipulation, but difficult for human understanding. The situation can be remedied by converting into polar notation with the standard relations: $Mag\,H(\omega) = [Re\,H(\omega)^2 + Im\,H(\omega)^2]^{1/2}$ and $Phase\,H(\omega) = \arctan[Re\,H(\omega) / Im\,H(\omega)]$. Working through the algebra